Lecture 11. International Agreements: Trade, Labor, and the Environment

Lectures on International Trade

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Outline

International Agreements

- International Trade Agreements
- International Agreements on Labor Issues
- International Agreements on the Environment

Introduction

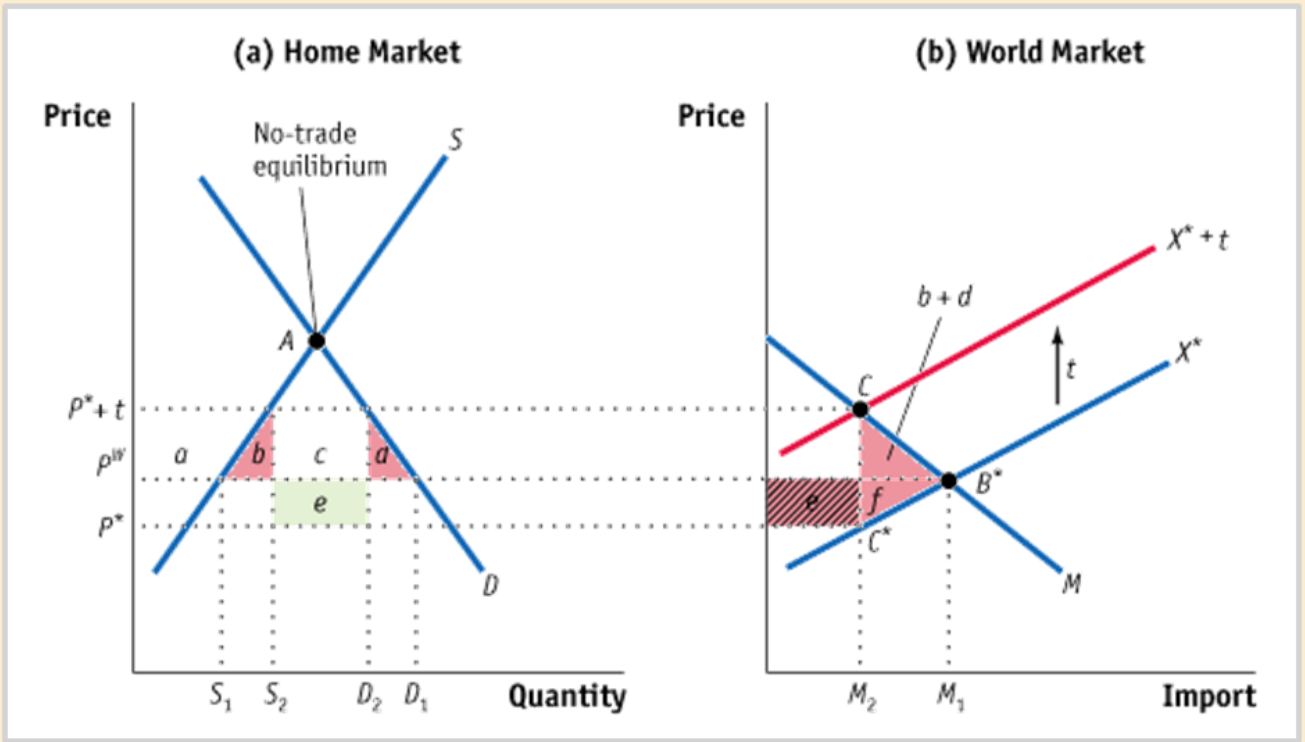
- In protests at the 1999 meeting of the World Trade Organization in Seattle, Washington, environmentalists dressed as turtles and other endangered species that had been affected by recent WTO rulings.
- Discussions about trade reform had been occurring since the formation of General Agreement on Trade and Tariffs (GATT) in 1947.
- However, never before had there been such an organized protest against it.
- The goal of this chapter is to examine why international agreements like those negotiated under the WTO for trade, and those negotiated for environmental reasons like the Copenhagen Summit, are needed.

International Trade Agreement

- By applying an import tariff: the tariff increases the import price for consumers in the large country but lowers the price received by foreign exporting firms.
- The reduction in the price received by exporters is a **terms-of-trade gain** for the importing country.
- When two or more countries apply tariffs against each other in an attempt to capture this terms-of-trade gain, they both end up losing.
- To avoid such losses, international agreements to reduce tariffs and move toward free trade are needed.
- The WTO is a **multilateral agreement**, involving many countries; There are also smaller **regional trade agreements**, involving several countries, often located near each other.

The Logic of Multilateral Trade Agreements

FIGURE 11-1 Tariff for a Large Country



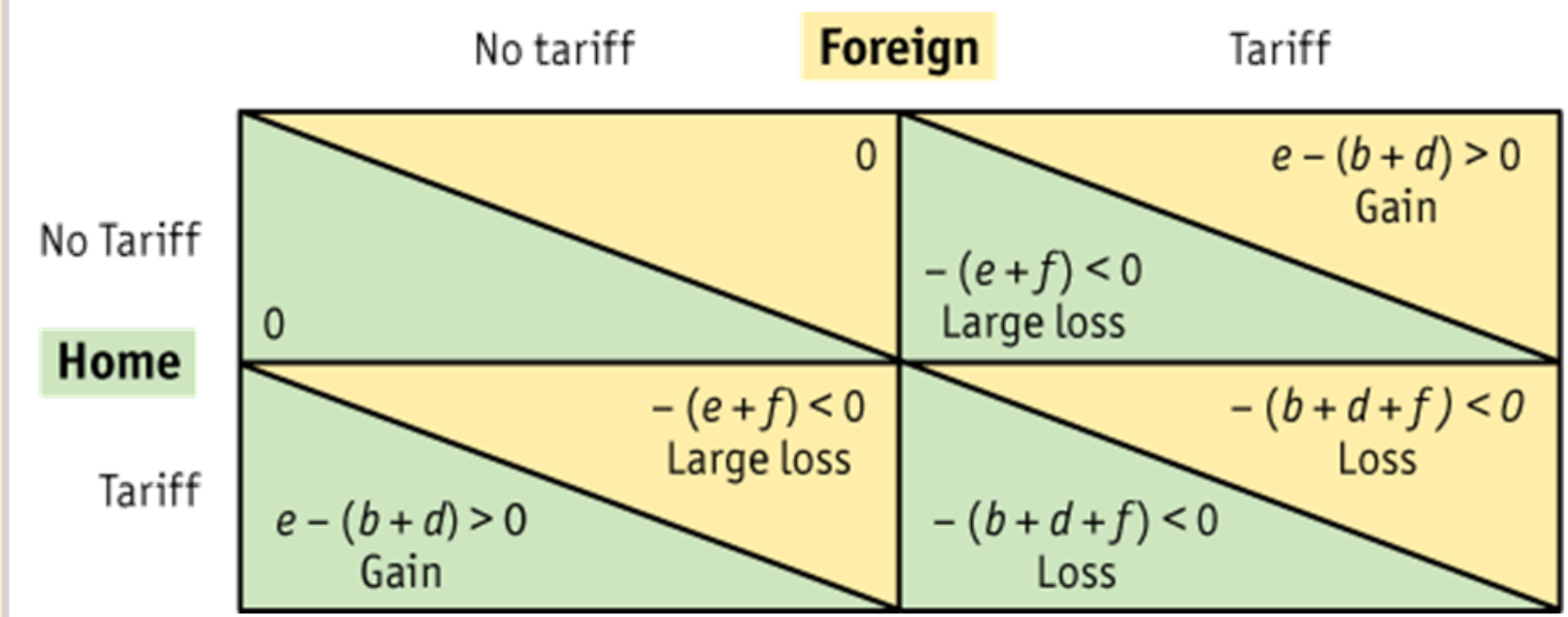
The deadweight loss at Home is the area of the triangle ($b + d$), and Home also has a terms-of-trade gain of area e.

Foreign loses the area ($e + f$), so the net loss in world welfare is the triangle ($b + d + f$).

International Trade Agreement, cont'

The Logic of Multilateral Trade Agreements

FIGURE 11-2 Payoffs in a Tariff Game



- This payoff matrix shows the welfare of the Home and Foreign countries as compared with free trade (upper-left quadrant in which neither country applies a tariff).

International Trade Agreement, cont'

- Welfare depends on whether one or both countries apply a tariff. The structure of payoffs is similar to the "prisoner's dilemma" because both countries suffer a loss when they both apply tariffs, and yet this is the unique Nash equilibrium.
- Each country acting on its own has an incentive to apply a tariff, but if they both apply tariffs, they will both be worse off.
- This bad outcome can be avoided if the countries enter into some kind of trade agreement.
- The WTO mechanism eliminated the prisoner's dilemma by providing an incentive to remove tariffs; the outcome was in the preferred upper-left quadrant of the payoff matrix in Figure 11-2, rather than the original Nash equilibrium in the lower-right quadrant.

International Trade Agreement, cont'

Regional Trade Agreements

- Under regional trade agreements, several countries eliminate tariffs among themselves but maintain tariffs against countries outside the region.
- Regional trade agreements are sometimes called **preferential trade agreements**, to emphasize that the member countries are favored over other countries.
- A **free-trade area** is a group of countries agreeing to eliminate tariffs (and other barriers to trade) among themselves but keeping whatever tariffs they formerly had with the rest of the world.
- A **customs union** is similar to a free-trade area, except that in addition to eliminating tariffs among countries in the union, the countries within a customs union also agree to a *common* schedule of tariffs with each country outside the union.

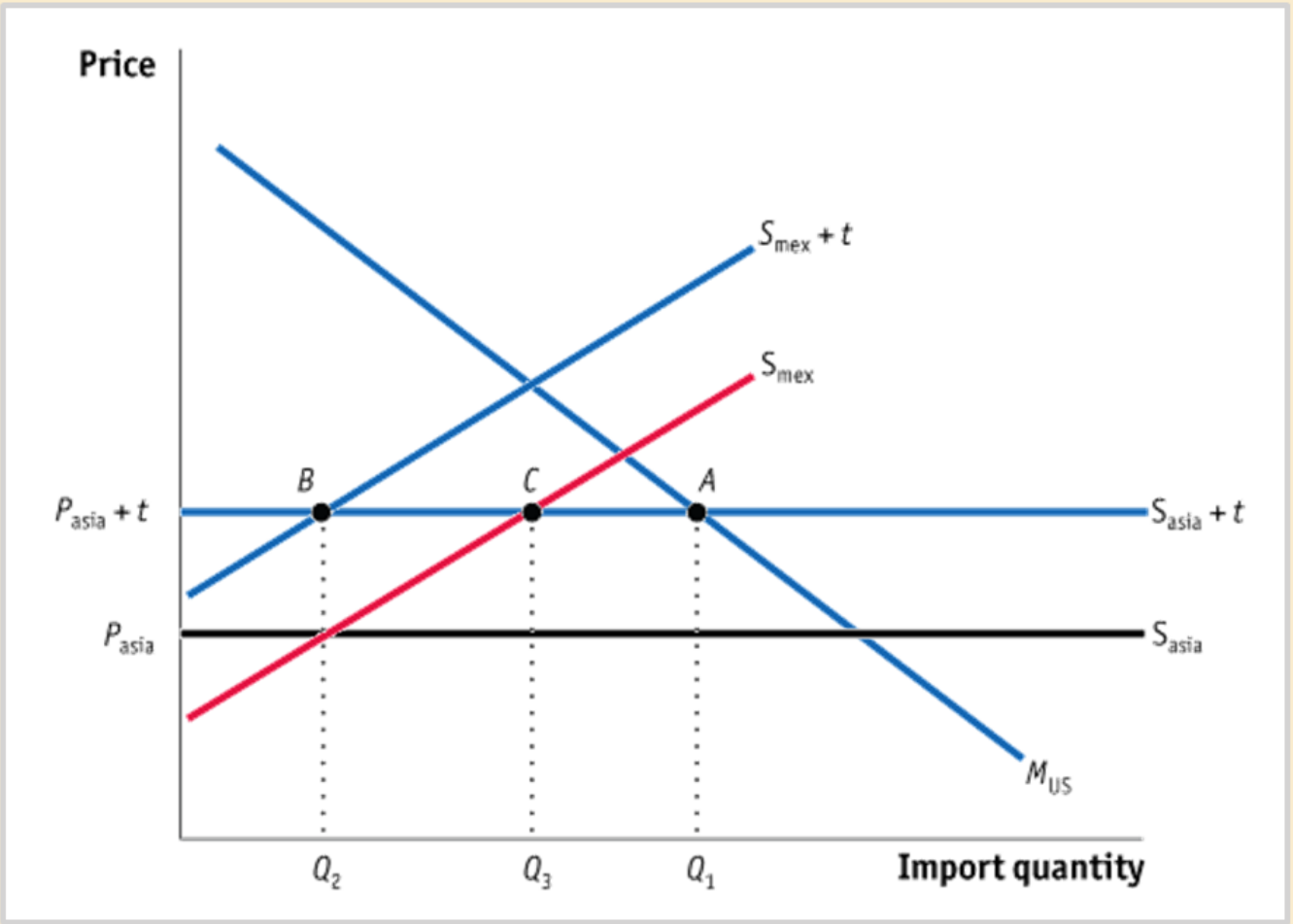
International Trade Agreement, cont'

Regional Trade Agreements

- Free-trade areas have complex **rules of origin**, which specify what type of goods can be shipped duty-free within the free-trade area. These rules are not needed in a customs union.
- When a regional trade agreement is formed and trade increases between member countries, the increase in trade can be of two types.
- The first type of trade increase, **trade creation**, occurs when a member country imports a product from another member country that formerly it produced for itself.
- The second reason for trade to increase within a regional agreement is due to **trade diversion**, which occurs when a member country imports a product from another member country that it formerly imported *from a country outside of the new trade region*.

Regional Trade Agreements

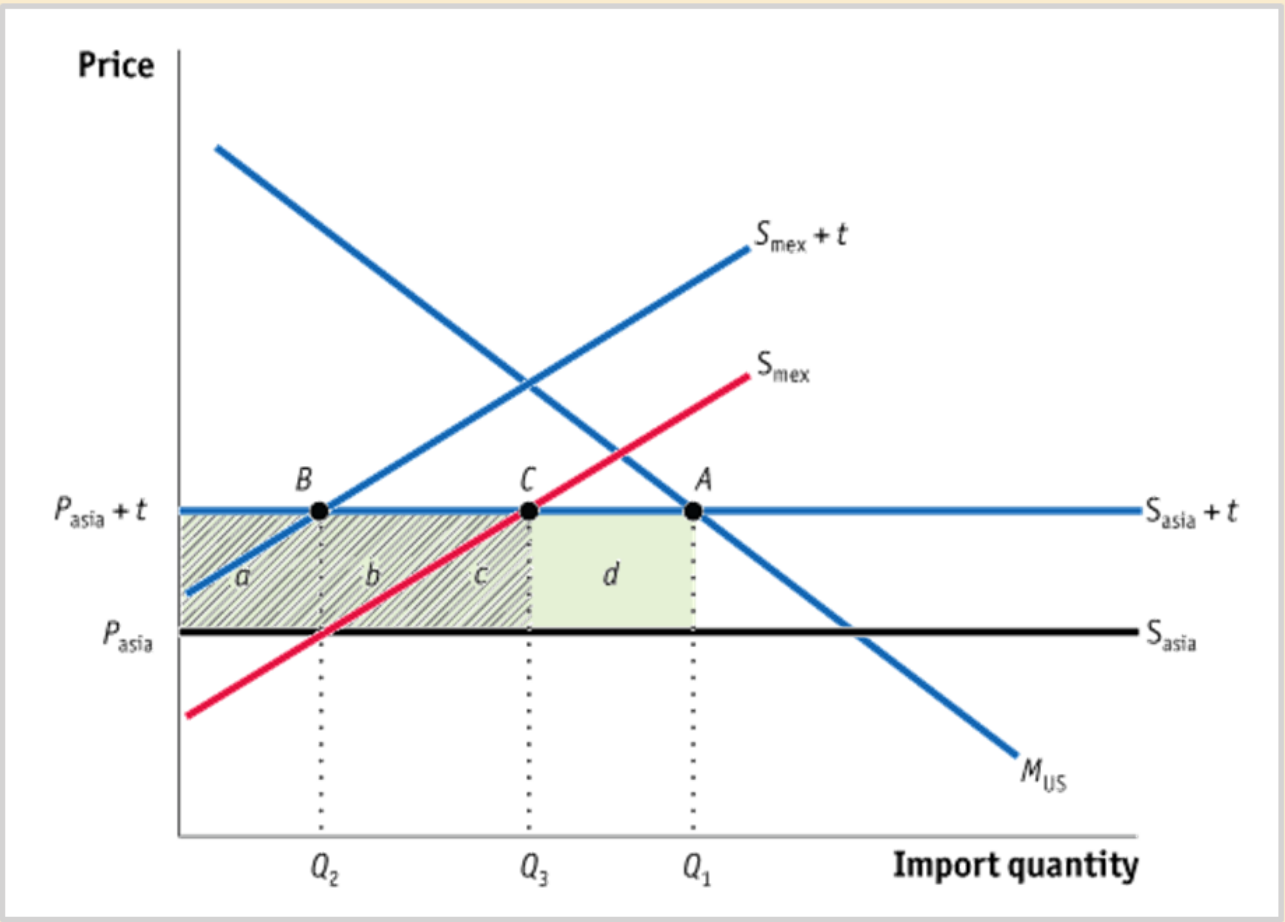
FIGURE 11-3 (1 of 2) **Trade Diversion**



With Mexico and Asia facing the same tariff of *t* for sales into the United States, the equilibrium is at *A* with the quantity Q_2 exported by Mexico and the remainder exported by Asia at a price of $P_{asia} + t$.

Regional Trade Agreements

FIGURE 11-3 (1 of 2) **Trade Diversion (continued)**



U.S. tariff revenue is the area ($a + b + c + d$).

Eliminating the tariff with Mexico under NAFTA leads to an expansion of Mexican exports to Q_3.

The United States loses the tariff revenue ($a + b + c$), which is the U.S. loss as a result of trade diversion from Asia to Mexico.

International Agreements on Labor Issues

- We use the term **labor standards** to refer to all issues that directly affect workers, including occupational health and safety, child labor, minimum wages, and so on.
- The labor side agreement negotiated under NAFTA does not change the existing labor laws in these countries but is meant to improve the enforcement of such laws.
- Besides the labor side agreement in NAFTA, there are many other examples of international agreements that monitor the conditions of workers in foreign countries.
- Unions and other organizations are concerned with issues such as job safety, the right of workers to unionize, workers' entitlement to breaks and not being forced to work overtime, and so on.

International Agreements on Labor Issues, cont'

TABLE 11-2 Survey Responses

This table summarizes the responses from a survey conducted by the National Bureau of Economic Research that asked about attitudes toward an item made under good working conditions and under poor working conditions.

Sample A	
Consumers who say they care about the condition of workers who make the clothing they buy:	
A lot	46%
Somewhat	38%
Only a little	8%
Not at all (or no response)	8%
Consumers willing to pay more for an item if assured it was made under good working conditions:	81%
Additional amount willing to pay for $10 item	$2.80
Additional amount willing to pay for $100 item	$15
Sample B	
Would choose an alternative to a T-shirt "with a nice logo" that local students say is made under poor working conditions if alternative is the same price	84%
Would not buy T-shirt made under poor working conditions at all	65%
Would buy T-shirt made under poor working conditions at average discount of	$4.30
Would pay more for T-shirt if came with assurance it was made under good conditions	78%
Average additional amount would pay (including as zeros those who did not offer to pay more or were inconsistent)	$1.83

International Agreements on Labor Issues, cont'

Other Labor Agreements

- **Corporate Responsibility** Because of the pressure from consumers and unions, corporations have started to monitor and improve the conditions in their overseas plants and the plants of their overseas subcontractors.
- **Country Responsibility** Several U.S. trade laws give the president the power to withhold trade privileges from countries that do not give their workers basic rights, including the right to organize.
- **Living Wage** Is it fair to expect foreign firms to pay a living wage to their workers, that is, a wage above the norm in the developing country? Economists have a ready answer: the wages should be as high as the market will allow, and not any higher.

International Agreements on Labor Issues, cont'

Wal-Mart Orders Chinese Suppliers to Lift Standards

- Wal-Mart, the world's biggest retailer, told its Chinese suppliers to meet strict environmental and social standards or risk losing its business.
- This was likely in response to growing criticism in the U.S. over issues that include labor conditions in its supplier factories
- The requirements include:
  - A clear demonstration of compliance with Chinese environmental laws
  - An improvement of 20 per cent in energy efficiency at the company's 200 largest China suppliers
  - Disclosure of the names and addresses of every factory involved in the production process

International Agreements on the Environment

- The WTO does not directly address environmental issues; other international agreements, called **multilateral environmental agreements**, deal specifically with the environment.
- Does Trade Help or Harm the Environment?
- The answer to the question of whether trade helps or harms the environment is that either outcome is possible, depending on the case being considered.
- **U.S. Sugar Quota** Producing ethanol from corn is much less energy-efficient than producing it from sugarcane.
- **U.S. Automobile VER** The "voluntary" export restraint (VER) on exports of Japanese cars to the United States, which began in 1981, limited the number of cars that Japanese firms could export each year, but not their value, so there was an incentive for the Japanese firms to export larger and/or more luxurious models. As the quality of the Japanese cars rose, so did the engine size and weight of the vehicles; as a result, the average gas mileage of the imported cars fell.

International Agreements on the Environment, cont'

TABLE 11-3 (1 of 2) **Environmental Cases at the GATT and WTO**

This table shows the outcome of environmental cases ruled upon by the GATT and WTO.

CASE	ISSUE	OUTCOME
Tuna-Dolphin		
In 1991 mexico appealed to the GATT against a U.S. ban on mexican tuna imports.	The United States put a ban on imports of tuna from Mexico that were not caught with nets that were safe for dolphins (as required in the united states under the Marine Mammal Protection Act).	In 1992 the gatt ruled in favor of Mexico that the U.S. import ban violated GATT rules. But the strong consumer response led to labeling of imported tuna as "dolphin friendly."
Shrimp-Turtle		
In 1996 India, Malaysia, Pakistan, and Thailand appealed to the WTO against a U.S. ban on shrimp imports.	The United States put a ban on imports of shrimp from India, Malaysia, Pakistan, and Thailand that were not caught with nets safe for sea turtles (as required in the United States under the Endangered Species Act).	In 1998 the WTO ruled in favor of India, Malaysia, Pakistan, and Thailand that the U.S. import ban violated WTO rules. But the United States could still require these countries to use turtle-safe nets, provided that adequate notice and consultation were pursued.

International Agreements on the Environment, cont'

TABLE 11-3 (2 of 2) **Environmental Cases at the GATT and WTO (continued)**

CASE	ISSUE	OUTCOME
Gasoline		
In 1994 Venezuela and Brazil appealed to the GATT against a U.S. ban on gasoline imports.	The United States put a ban on imports of gasoline from Venezuela and Brazil because the gas exceeded the maximum amount allowed of a smog-causing chemical (under the U.S. Clean Air Act).	In 1996 the WTO ruled in favor of Venezuela and Brazil that the U.S. import restriction violated equal treatment of domestic and foreign producers. The United States adjusted the rules to be consistent with the WTO and still pursued its own clean air goals.
Biotech Food		
In 2003 the United States appealed to the WTO that Europe was keeping out genetically modified food and crops.	Since 1998 no imports of genetically modified food or crops had been approved in the European Union.	In 2006 the WTO ruled that the European actions violated the principle that import restrictions must be based on "scientific risk assessments." Labeling and consumer concerns in Europe will still limit such imports.

International Agreements on the Environment, cont'

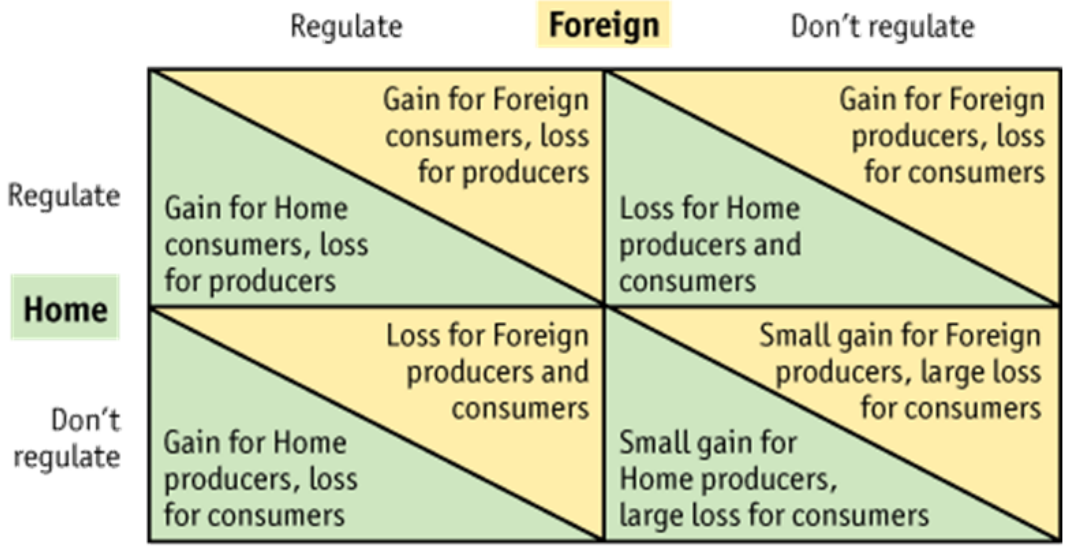
Trade in Fish What has led to overharvesting of fish?

- Economists believe that this outcome occurs whenever people are competing for the same resource stock (fish, in this case) and refer to this phenomena as the **tragedy of the commons**.
- When a resource such as fish is treated as **common property** that anyone can harvest, it will be subject to overfishing and its stocks will diminish rapidly over time as each producer seeks to maximize its own share of the resource.
- The Solution to the Tragedy of the Commons: If instead there was a system of rules that assigned property rights to the resource and limited the harvest of each person, then the overharvesting could be avoided.
- When a resource is truly global in nature, however, then it is not enough to assign property rights within a country. International agreements for fish and other endangered species are arranged through the Convention on International Trade in Endangered Species (CITES).

International Agreements on the Environment, cont'

International Agreements on Pollution

FIGURE 11-6 **Payoffs in an Environmental Game**



This payoff matrix shows the gains and losses for Home and Foreign countries, depending on whether they adopt environmental regulations. If governments weight producer surplus more than consumer surplus, then the structure of payoffs is similar to the prisoner's dilemma because the Nash equilibrium is to have both countries not adopt regulations. That outcome can occur with "global" pollutants.

Multilateral Agreements One example of an international agreement is the Montreal Protocol on Substances That Deplete the Ozone Layer, which has successfully eliminated the use of chlorofluorocarbons (CFCs).

International Agreements on the Environment, cont'

The Kyoto Protocol and the Copenhagen Accord

- The **Kyoto Protocol** built on the United Nations' 1992 treaty on climate change, established specific targets for reduction in greenhouse gas emissions: the industrial countries should cut their emissions of greenhouse gases by a collective 5.2% less than their 1990 levels.
- There are four reasons often given to explain why the United States did not join the Kyoto Protocol:
  - although the evidence toward global warming is strong, we still do not understand all the consequences of policy actions;
  - while the United States is the largest emitter of greenhouse gases, meeting the Kyoto targets would negatively affect its economy;
  - Kyoto failed to include the developing countries, especially China and India;
  - there are other ways to pursue reductions in greenhouse gas emissions.

International Agreements on the Environment, cont'

The Kyoto Protocol and the Copenhagen Accord

- The **Copenhagen Accord** is a recognition that further increases in global average temperature should be kept below 2 degrees centigrade.
- Under this accord, an agreement was made that industrialized countries will submit goals for greenhouse gas emissions reductions, while developing countries will contribute their efforts in this regard.
- There is also the establishment of a fund to finance the needs of developing countries in fighting the effect of climate change.

Summary

Key Points:

1. There are two primary types of free-trade agreements: multilateral and regional. Multilateral agreements are negotiated among large groups of countries (such as all countries in the WTO) to reduce trade barriers among them, whereas regional agreements operate among a smaller group of countries, often in the same region.

2. Under perfect competition, we can analyze the benefits of multilateral agreements by considering the Nash equilibrium of a two-country game in which the countries are deciding whether to apply a tariff. The unique Nash equilibrium for two large countries is to apply tariffs against each other, which is an example of a "prisoner's dilemma". By using an agreement to remove tariffs, both countries become better off by eliminating the deadweight losses of the tariffs.

Summary

Key Points:

3. Regional trade agreements are also known as "preferential trade agreements", because they give preferential treatment (i.e., free trade) to the countries included within the agreement, but maintain tariffs against outside countries. There are two types of regional trade agreements: free-trade areas (such as NAFTA) and customs unions (such as the European Union).
4. The welfare gains and losses that arise from regional trade agreements are more complex than those that arise from multilateral trade agreements because only the countries included within the agreement have zero tariffs, while tariffs are maintained against the countries outside the agreement. Under a free-trade area, the countries within the regional trade agreement each have their own tariffs against outside countries; whereas under a customs union, the countries within the regional trade agreement have the same tariffs against outside countries.

Summary

Key Points:

5. Trade creation occurs when a country within a regional agreement imports a product from another member country that formerly it produced for itself. In this case, there is a welfare gain for both the buying and the selling country.

6. Trade diversion occurs when a member country imports a product from another member country that it formerly imported from a country outside of the new trade region. Trade diversion leads to losses for the former exporting country and possibly for the importing country and the new trading region as a whole.